Enrolment No	:	**2400636650**
Name	:	**Nikhil Kumar**
Address	:	**Partap singh house near pole no 352 village chhawla new delhi-110071**

Course Title	:	Computer Networks and Data Mining Lab
Course Code	:	**MCSL-223**
Assignment No	:	MCA_NEW(II)/L-223/Assignment/24-25
Study Center	:	**Rajdhani College** **Raja Garden, LSC-38046**
Phone No	:	**8802480960**
Email ID	:	**n.shokeen0007@gmail.com**
Date	:	02/11/24
Signature	:	

इंदिरा गांधी राष्ट्रीय मुक्त विश्वविद्यालय
मैदान गढ़ी, नई दिल्ली - 110068

Indira Gandhi National Open University
Maidan Garhi,New Delhi - 110068

IGNOU - Student Identity Card

2400636650

Enrolment Number :	**2400636650**
RC Code :	**38: DELHI 3 Naraina**
Name of the Programme :	**MCA_NEW : Master of Computer Applications**
Name :	**NIKHIL KUMAR**
Guardian's Name :	**JITENDER SINGH**
Address :	**Partap singh house , near pole no 352 village chha, SOUTH WEST**
Pin Code :	**110071**

Instructions:
1. This card should be produced on demand at the Study Center, Examination Center or any other Establishment of IGNOU to use its facilities.
2. The facilities would be available only relating to the Programme/course for which the student is registered.
3. This ID Card is generated online. Students are advised to take a color print of this ID Card and get it laminated.
4. The student details can be cross checked with the QR Code at www.ignou.ac.in. PD 2024-04-30

Registrar
Student Registration Division

Page No.
Date:

MCS = 223

Computer Networks And Data Mining Lab

Part = 1

Q-1 setup 4 nodes two tcp client and server pair and two UDP client and server pair. send packets respective.

Ans To set up a network with 4 nodes consiting of two client pai and two udp client - server.

Page No.
Date:

traffic to plot the number of
we'll follow steps.

(1) Setup the network nodes :-

(i) Node → Tcp server.
(ii) Node → Tcp client.
(iii) Node → UDP server.
(iv) Node → UDP client.

(2) establish connections :-

(i) Tcp server (Node 1) communicati with Tcp client (Node)
(ii) UDP server (Node 3) communication with client.

(3) Send Packets :-

(i) Tcp server sends packets to Tcp client.)
(ii) UDP server sends packet to client.

(4) monitor traffic :- Traffic

Page No.
Date:

number of bytes received each client

□ Assumption :-

(I) We assume that the servers are running on different IP adress.

(II) We assumed that the client can research the server and there are no network issue.

(III) We assumed that the traffic monitoring tool can capture nessacry date.

□ Implementation :-

we'll create python script to simulate this setup we'll use the socket library for network communication plotting the all result.

(1) Step 1 :-

Tcp server and

Page No.
Date:

client
Tcp server (Node)
important soket

□ def tcp_server (i) :-

server_socket = socket.socket
(socket.AF_NET, socket.sock
stream)
server_socket.bind (localhost)
server_socket.listen (1)
Print ("Tcp server is listeni
ng on port 8080 ...")

Conn, addr = server_socket
Print (Connected by ladder)
while true,
Conn.sendall (b" Hello Tcp cli
ent).

step 2 :-
UDP server and cli
ent.
UDP server
important soket.

Page No.
Date:

The male female ratio Admit Card allotment.
This is for comes certain musti tution admit Card to either the malred or the females.

- Institute Analyse

Tems of similar as the report generation. The estintially needed when it comes to grating online Admit card system.
The keyword in the report consists basis Admit Card produce like Admit Card.

- (1) student study in Regional center

many student in one Regional center but one student can not one student many Regionar center.

(II) Coordinator Controls a Regional

Page No.
Date:

Center :-

student study under the Coordinator of the Regional center and the coordinate records the date of admission of student in the Regional center.

(iii) Regional center has many Course :-

One Regional center has many Course but one Course can not come under many Regional center.

(iv) Students study on Course :-

One student study on several Course and the number of hours study by the student on single Course is record.
The student several Course.

Page No.
Date:

□ Examplation :-

(I) Tcp and UDP server - client Implementation.
(II) Traffic monitoring
(III) Plotting the data.

(I) Tcp and UDP server - client Implentation :-

(I) The Tcp server listens on port 8080 and sends "Hellow tcp client massage.
(II) The tcp client Connects to the tcp server receives mess and Contact the byte 8028
(III) The UDP client listen port 8028 receive massage and Counts byte received.

(II) Traffic monitoring :-

(I) A separte thread monitor traffic tracks the bytes.

Page No.
Date:

(II) This data is stored in tcp_received_byte and udp_received_byte list.

(3) Plotting the Data :-

(1) After monitoring for 60 second the plot_traffic function is called to plot the number of type received by both udp clients.

Part II

Q-1 Implement simple k-means algorithm to demonstrate the clustering rule on the following datasets.

Ans This experiment illustanced the use of simple k-mean clusting with weka explonder.
The sample data set used for the example is based on the iris data available.
This document assumes that

Page No.
Date:

```
def _ udp_ server
server _ socket = socket.
server _ bind (llocahist, 8081)
  Print (UDP server listering)
while true
server _ socket (b" Hello UDP
  client (localhost, 8089)
```

- **Received _ bytes** ⇒

```
data, addr = client _ socket . re
  cvfrom (1024)
received _ bytes + = (data)
Print (f" Received {len(data)}
except keyboardterrupt
Print ("Totall bytes received:
  {(received _ bytes)}
```

- **Step 3** :- Monitoring and plo
ttig.

Important matplotlib . pylot as plt
Important time.
Important threading.
Tcp _ received _ bytes.

Page No.
Date:

□ def plot_traffic :

(i) Plt. figure (figsize =(10,5))
(ii) Plt. plot (tcp_received_byyet
label = Tcp bytes Received)
(iii) Plt. plot (udp_received_byte
label = UDP bytes Received)
(iv) Plt.xlable (time (seconds))
(v) Plt.ylable (byte Received)
(vi) Plt title (byte)

□ If __name__ = __main__ :

(i) Tcp_server_thead = theating.
thread (target = tcP_server)

(ii) udp_server_thread = theading
(target = udp_server)
(iii) Tcp_server_thread start()
(iv) Tcp client_thread.start ()
(v) udp_server_thread.start
(vi) udp_client_thread.start()
(vii) Moniter_traffic_thread =
theading.
(viii) monitor_traffic_theads.

Page No.
Date:

This document assumes that pre process has been perform This dis least include 150 instanse.

(1) Step 1 :- Run the weka explore and load the data life aris preparassing interface.

(2) Step 2 :- Inorder to perfo clusting select the 'cluser' tab in the explore and click on the choose button.

(3) Step 3 :- In this case we selected simple K-means.

(4) Step 4 :- Next click text button to the right of the choose button to get pre pup.

window shown in the screen in this we enter six on the number of cluster and we leave the value of the seeds on set it is.

(5) Step 5 :-

Once of the optio have been specified. we run the clusting algorithm there we must make sure they are in the mode panel.

(6) Step 6 :-

The result window shows the screenshort. Here cluster centional are mea vectors for each cluster. This cluster used to centroid of shows the class.

(7) Step 7 :-

Another way of under stand of each cluster through we can do it try right the

result set on the result. List pannel and selecting the cluster assignment.

- Indors to perform clusting the cluser's tab in the explore window shown in the screenshot we enter six on the number of clusters and we leave the value of the head.

- Next click in text button to the right of the choose button to get popup window shown in the screenshort. In this we enter six on the number of and we leave.

- Once of the option have been specified. We run the we must make sure that they are mode panel. The use of trainning set option is selected. This process and resulting.

Page No.
Date:

- Another way of understanding of each cluster through we can do this try right click the result set on result list.

Choose

(i) use a aging set
(ii) suplied test set
(iii) Percentage
(iv) Classes of evalution.

[(Nom) class]

[Start] [stop]

... Run Information ...

Scheme : meta stop -1
Relation : itra
Instances : 150
Attribute : 5
class
Petalwidth
Seeplation

Teste code : Qualuese on trainning date

= model and evualtion set ... — —

Page No.
Date:

Interpretation of the above visualization :-

From the above visualization we can understand the distriction of spaci lenghth and patel length in each cluster for insta for each culster is domi nated patel.

Step 8 :-

we can assure that resulting datest which include each instance along its assign culster.
To do so we click the save button in the valiza tion window and save the result.
Iristr means the stop fro ntation of this life is shown in the following The following screenshort fo clustring.

Page No.
Date:

	choose	$H^2 - A$ weka last 1to
	cluster mode.	
(i)	use trainning to set.	= = = Run Information = = =
(ii)	Supplied test set	Scheme :- weka
(iii)		classers slaptren
(iv)	Percentage splite,	weka.
(iv)	class to dusters	Relation :- tbuk
(v)	(None) Lansup	Instrance :- 14
		Atribute :- 5
(vi)	☑ store duster for visulation	age
		Income
		student
(vii)	Ignore atri	buyye
(viii)	Start	= = = = = = =
		Age <30 <30
(ix)	12:96:95	Income medium med
(x)	12:27:39:	student use no
	Simple mark	< >